

5. Nuclear medicine involves the use of radioactive substances in the diagnosis and treatment of disease. Nuclear medicine records radiation emitted from within the body rather than radiation that is generated by external sources like X-rays. Nuclear medicine scans differ from radiology as the emphasis is not on imaging but on organ function.

Information about some isotopes that emit gamma (γ) rays is given in the table.

Radioisotope	Half-life	Energy of γ rays
caesium-137	30.17 years	0.662 MeV
cobalt-60	5.26 years	1.17 MeV
iodine-125	59.6 days	31.4 keV
thallium-201	73.0 hours	71.0 keV
palladium-103	17.0 days	21.0 keV

- (a) (i) Describe the nature of a gamma (γ) ray.

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- (ii) Explain why radioisotopes that emit gamma rays are suitable for examining organ function when injected into the body.

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- (iii) Explain which is the most suitable radioisotope from the table to inject into the human body as part of a gamma camera investigation.

[2]

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- (b) In order to treat prostate cancer, palladium-103 pellets are placed directly into the prostate gland. They remain permanently in place. The gamma emissions from the pellets are almost undetectable when their activity drops to $\frac{1}{32}$ of its original value. Calculate the time taken for this reduction to occur. [4]

Time = days

END OF PAPER

Examiner
only

10

THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1										4 He Helium 2																									
7 Li Lithium 3		9 Be Beryllium 4		11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10																					
23 Na Sodium 11		24 Mg Magnesium 12		27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18																					
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		127 I Iodine 53		131 Xe Xenon 54			
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															
														Key																					

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number